Topic: Least squares solution

Question: Find the least squares solution to the system.

$$x - 2y = 5$$

$$3x + y = -6$$

$$-x - 2y = -2$$

Answer choices:

A $\vec{x}^* = \left(-\frac{9}{10}, -\frac{11}{10}\right)$

B $\vec{x}^* = \left(\frac{7}{10}, \frac{11}{10}\right)$

C $\vec{x}^* = (-11, -12)$

D $\vec{x}^* = \left(-\frac{7}{10}, -\frac{11}{10}\right)$

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Solution: D

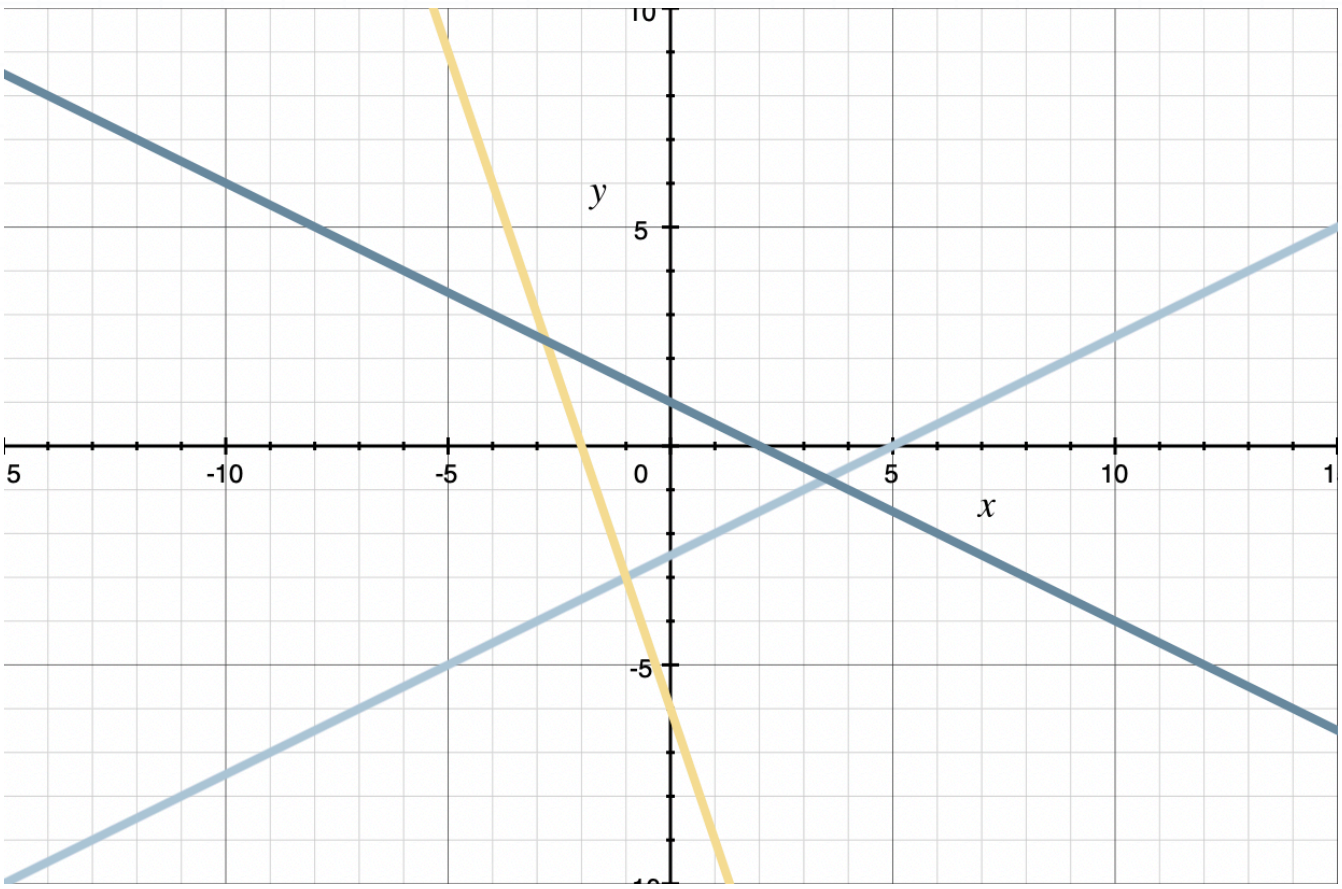
Put each line into slope-intercept form,

$$y = \frac{1}{2}x - \frac{5}{2}$$

$$y = -3x - 6$$

$$y = -\frac{1}{2}x + 1$$

then graph all three in the same plane.



While there are two points where some of the lines intersect, there's no single point where all three lines intersect, which means there's no solution to $A\vec{x} = \vec{b}$.

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$$\begin{bmatrix} 1 & -2 \\ 3 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ -6 \\ -2 \end{bmatrix}$$

In other words, $\vec{b} = (5, -6, -2)$ is not in the column space of the coefficient matrix A, and there's no vector $\vec{x} = (x, y)$ you can find that makes the $A\vec{x} = \vec{b}$ equation true.

The next best thing we can do is find the least squares solution. By building the matrix equation, we've already found

$$A = \begin{bmatrix} 1 & -2 \\ 3 & 1 \\ -1 & -2 \end{bmatrix}$$

Now we'll find A^T.

$$A^T = \begin{bmatrix} 1 & 3 & -1 \\ -2 & 1 & -2 \end{bmatrix}$$

Then $A^T A$ is

$$A^T A = \begin{bmatrix} 1 & 3 & -1 \\ -2 & 1 & -2 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 3 & 1 \\ -1 & -2 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1(1) + 3(3) - 1(-1) & 1(-2) + 3(1) - 1(-2) \\ -2(1) + 1(3) - 2(-1) & -2(-2) + 1(1) - 2(-2) \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 + 9 + 1 & -2 + 3 + 2 \\ -2 + 3 + 2 & 4 + 1 + 4 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 11 & 3 \\ 3 & 9 \end{bmatrix}$$

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And $A^T\overrightarrow{b}$ is

$$A^T\overrightarrow{b} = \begin{bmatrix} 1 & 3 & -1 \\ -2 & 1 & -2 \end{bmatrix} \begin{bmatrix} 5 \\ -6 \\ -2 \end{bmatrix}$$

$$A^T\overrightarrow{b} = \begin{bmatrix} 1(5) + 3(-6) - 1(-2) \\ -2(5) + 1(-6) - 2(-2) \end{bmatrix}$$

$$A^T\overrightarrow{b} = \begin{bmatrix} 5 - 18 + 2 \\ -10 - 6 + 4 \end{bmatrix}$$

$$A^T\overrightarrow{b} = \begin{bmatrix} -11 \\ -12 \end{bmatrix}$$

Then we get

$$A^TA\overrightarrow{x}^* = A^T\overrightarrow{b}$$

$$\begin{bmatrix} 11 & 3 \\ 3 & 9 \end{bmatrix} \overrightarrow{x}^* = \begin{bmatrix} -11 \\ -12 \end{bmatrix}$$

Then to find $\overrightarrow{x}^*$, we'll put the augmented matrix into reduced row-echelon form.

$$\begin{bmatrix} 11 & 3 & | & -11 \\ 3 & 9 & | & -12 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \frac{3}{11} & | & -1 \\ 3 & 9 & | & -12 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \frac{3}{11} & | & -1 \\ 0 & \frac{90}{11} & | & -9 \end{bmatrix}$$

$$\begin{bmatrix} 1 & \frac{3}{11} & | & -1 \\ 0 & 1 & | & -\frac{11}{10} \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & | & -\frac{7}{10} \\ 0 & 1 & | & -\frac{11}{10} \end{bmatrix}$$

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Then the least squares solution is given by the augmented matrix as

$$\overrightarrow{x}^* = \left(-\frac{7}{10}, -\frac{11}{10} \right)$$

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Topic: Least squares solution

Question: Find the least squares solution to the system.

$$3x - 2y = -6$$

$$x - 5y = -5$$

$$x + y = 4$$

Answer choices:

A $\vec{x}^* = \left(-\frac{16}{23}, \frac{261}{230}\right)$

B $\vec{x}^* = \left(\frac{261}{230}, -\frac{16}{23}\right)$

C $\vec{x}^* = \left(\frac{16}{23}, -\frac{261}{230}\right)$

D $\vec{x}^* = \left(-\frac{261}{230}, \frac{16}{23}\right)$

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Solution: A

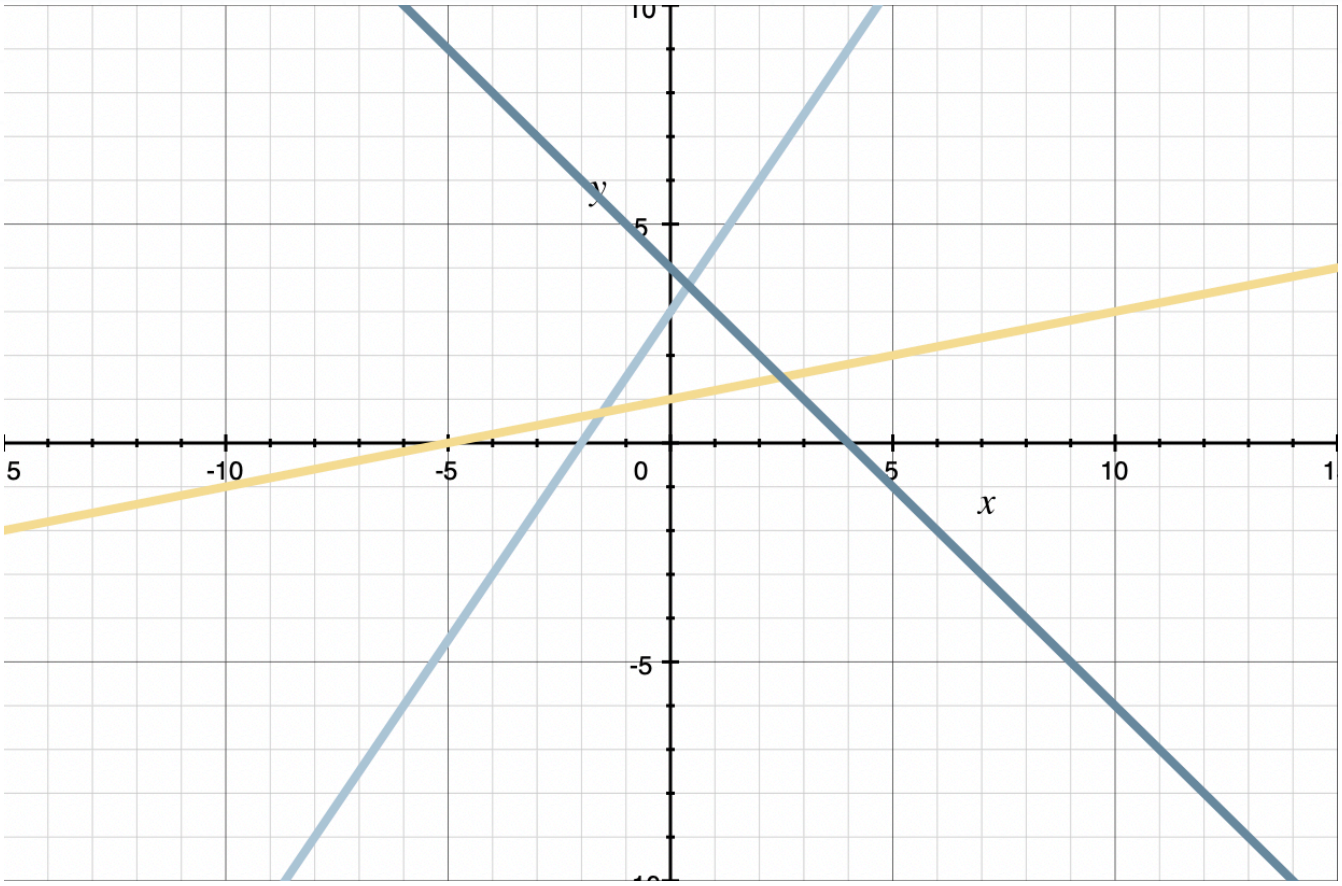
Put each line into slope-intercept form,

$$y = \frac{3}{2}x + 3$$

$$y = \frac{1}{5}x + 1$$

$$y = -x + 4$$

then graph all three in the same plane.



While there are three points where some of the lines intersect, there's no single point where all three lines intersect, which means there's no solution to $A\vec{x} = \vec{b}$.

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$$\begin{bmatrix} 3 & -2 \\ 1 & -5 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -6 \\ -5 \\ 4 \end{bmatrix}$$

In other words, $\vec{b} = (-6, -5, 4)$ is not in the column space of the coefficient matrix A, and there's no vector $\vec{x} = (x, y)$ you can find that makes the $A\vec{x} = \vec{b}$ equation true.

The next best thing we can do is find the least squares solution. By building the matrix equation, we've already found

$$A = \begin{bmatrix} 3 & -2 \\ 1 & -5 \\ 1 & 1 \end{bmatrix}$$

Now we'll find A^T.

$$A^T = \begin{bmatrix} 3 & 1 & 1 \\ -2 & -5 & 1 \end{bmatrix}$$

Then $A^T A$ is

$$A^T A = \begin{bmatrix} 3 & 1 & 1 \\ -2 & -5 & 1 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ 1 & -5 \\ 1 & 1 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 3(3) + 1(1) + 1(1) & 3(-2) + 1(-5) + 1(1) \\ -2(3) - 5(1) + 1(1) & -2(-2) - 5(-5) + 1(1) \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 9 + 1 + 1 & -6 - 5 + 1 \\ -6 - 5 + 1 & 4 + 25 + 1 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 11 & -10 \\ -10 & 30 \end{bmatrix}$$

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And $A^T\vec{b}$ is

$$A^T\vec{b} = \begin{bmatrix} 3 & 1 & 1 \\ -2 & -5 & 1 \end{bmatrix} \begin{bmatrix} -6 \\ -5 \\ 4 \end{bmatrix}$$

$$A^T\vec{b} = \begin{bmatrix} 3(-6) + 1(-5) + 1(4) \\ -2(-6) - 5(-5) + 1(4) \end{bmatrix}$$

$$A^T\vec{b} = \begin{bmatrix} -18 - 5 + 4 \\ 12 + 25 + 4 \end{bmatrix}$$

$$A^T\vec{b} = \begin{bmatrix} -19 \\ 41 \end{bmatrix}$$

Then we get

$$A^T A\vec{x}^* = A^T\vec{b}$$

$$\begin{bmatrix} 11 & -10 \\ -10 & 30 \end{bmatrix} \vec{x}^* = \begin{bmatrix} -19 \\ 41 \end{bmatrix}$$

Then to find $\vec{x}^*$, we'll put the augmented matrix into reduced row-echelon form.

$$\begin{bmatrix} 11 & -10 & | & -19 \\ -10 & 30 & | & 41 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{10}{11} & | & -\frac{19}{11} \\ -10 & 30 & | & 41 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{10}{11} & | & -\frac{19}{11} \\ 0 & \frac{230}{11} & | & \frac{261}{11} \end{bmatrix}$$

$$\begin{bmatrix} 1 & -\frac{10}{11} & | & -\frac{19}{11} \\ 0 & 1 & | & \frac{261}{230} \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & | & -\frac{16}{23} \\ 0 & 1 & | & \frac{261}{230} \end{bmatrix}$$

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Then the least squares solution is given by the augmented matrix as

$$\vec{x}^* = \left(-\frac{16}{23}, \frac{261}{230} \right)$$

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Topic: Least squares solution

Question: Find the least squares solution to the system.

$$x + 2y = -4$$

$$x - y = 3$$

$$y = 2$$

Answer choices:

A $\vec{x}^* = \left(\frac{3}{11}, -\frac{17}{11}\right)$

B $\vec{x}^* = \left(-\frac{3}{11}, \frac{17}{11}\right)$

C $\vec{x}^* = \left(\frac{17}{11}, -\frac{3}{11}\right)$

D $\vec{x}^* = \left(-\frac{17}{11}, \frac{3}{11}\right)$

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Solution: A

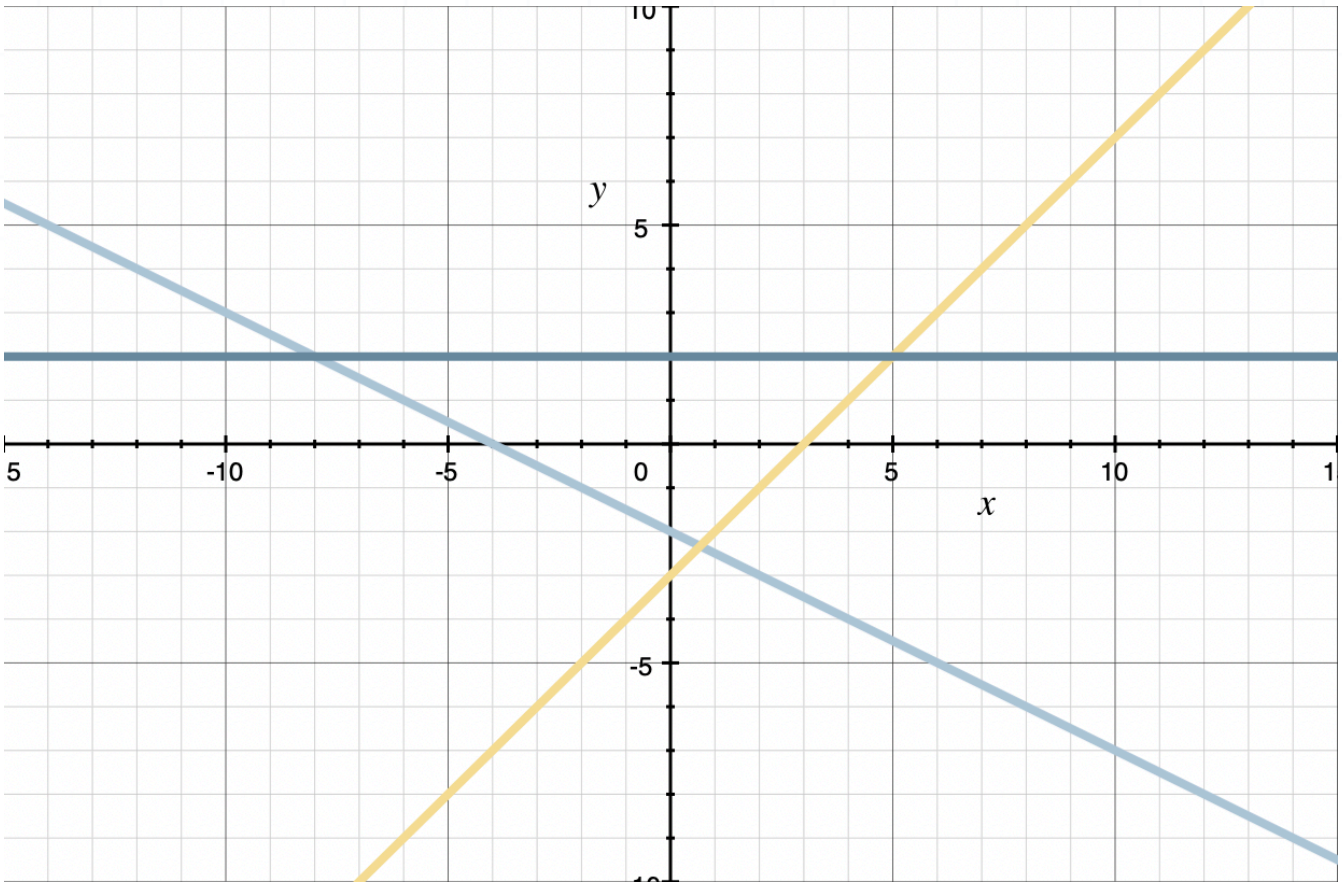
Put each line into slope-intercept form,

$$y = -\frac{1}{2}x - 2$$

$$y = x - 3$$

$$y = 2$$

then graph all three in the same plane.



While there are three points where some of the lines intersect, there's no single point where all three lines intersect, which means there's no solution to $A\vec{x} = \vec{b}$.

$$\begin{bmatrix} 1 & 2 \\ 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -4 \\ 3 \\ 2 \end{bmatrix}$$

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In other words, $\overrightarrow{b} = (-4,3,2)$ is not in the column space of the coefficient matrix A, and there's no vector $\overrightarrow{x} = (x, y)$ you can find that makes the $A\overrightarrow{x} = \overrightarrow{b}$ equation true.

The next best thing we can do is find the least squares solution. By building the matrix equation, we've already found

$$A = \begin{bmatrix} 1 & 2 \\ 1 & -1 \\ 0 & 1 \end{bmatrix}$$

Now we'll find A^T.

$$A^T = \begin{bmatrix} 1 & 1 & 0 \\ 2 & -1 & 1 \end{bmatrix}$$

Then $A^T A$ is

$$A^T A = \begin{bmatrix} 1 & 1 & 0 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1(1) + 1(1) + 0(0) & 1(2) + 1(-1) + 0(1) \\ 2(1) - 1(1) + 1(0) & 2(2) - 1(-1) + 1(1) \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 + 1 + 0 & 2 - 1 + 0 \\ 2 - 1 + 0 & 4 + 1 + 1 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 2 & 1 \\ 1 & 6 \end{bmatrix}$$

And $A^T\overrightarrow{b}$ is

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$$A^T\vec{b} = \begin{bmatrix} 1 & 1 & 0 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} -4 \\ 3 \\ 2 \end{bmatrix}$$

$$A^T\vec{b} = \begin{bmatrix} 1(-4) + 1(3) + 0(2) \\ 2(-4) - 1(3) + 1(2) \end{bmatrix}$$

$$A^T\vec{b} = \begin{bmatrix} -4 + 3 + 0 \\ -8 - 3 + 2 \end{bmatrix}$$

$$A^T\vec{b} = \begin{bmatrix} -1 \\ -9 \end{bmatrix}$$

Then we get

$$A^T A\vec{x}^* = A^T\vec{b}$$

$$\begin{bmatrix} 2 & 1 \\ 1 & 6 \end{bmatrix} \vec{x}^* = \begin{bmatrix} -1 \\ -9 \end{bmatrix}$$

Then to find $\vec{x}^*$, we'll put the augmented matrix into reduced row-echelon form.

$$\begin{bmatrix} 2 & 1 & | & -1 \\ 1 & 6 & | & -9 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 6 & | & -9 \\ 2 & 1 & | & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 6 & | & -9 \\ 0 & -11 & | & 17 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 6 & | & -9 \\ 0 & 1 & | & -\frac{17}{11} \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & | & \frac{3}{11} \\ 0 & 1 & | & -\frac{17}{11} \end{bmatrix}$$

Then the least squares solution is given by the augmented matrix as

$$\vec{x}^* = \left(\frac{3}{11}, -\frac{17}{11} \right)$$

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